

Tutorial 7 Answers

Problem 8-1 and Problem 8-4

Use teaching slides as a reference to sketch and get the answers

Problem 8-46

Rise $\beta_1 = 75^\circ$ $h_1 = 50 \text{ mm}$	Dwell $\beta_2 = 75^\circ$ $h_2 = 0$	Fall $\beta_3 = 75^\circ$ $h_3 = 50 \text{ mm}$	Dwell $\beta_4 = 135^\circ$ $h_4 = 0$
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Using 4-5-6-7 polynomial.

$$\begin{aligned}
 S(\theta, \beta, h) &= h \left[35 \left(\frac{\theta}{\beta} \right)^4 - 84 \left(\frac{\theta}{\beta} \right)^5 + 70 \left(\frac{\theta}{\beta} \right)^6 - 20 \left(\frac{\theta}{\beta} \right)^7 \right] \\
 \left\{ \begin{aligned}
 V(\theta, \beta, h) &= \frac{h}{\beta} \left[140 \left(\frac{\theta}{\beta} \right)^3 - 420 \left(\frac{\theta}{\beta} \right)^4 + 420 \left(\frac{\theta}{\beta} \right)^5 - 140 \left(\frac{\theta}{\beta} \right)^6 \right] \\
 A(\theta, \beta, h) &= \frac{h}{\beta^2} \left[420 \left(\frac{\theta}{\beta} \right)^2 - 1680 \left(\frac{\theta}{\beta} \right)^3 + 2100 \left(\frac{\theta}{\beta} \right)^4 - 840 \left(\frac{\theta}{\beta} \right)^5 \right] \\
 J(\theta, \beta, h) &= \frac{h}{\beta^3} \left[840 \left(\frac{\theta}{\beta} \right) - 5040 \left(\frac{\theta}{\beta} \right)^2 + 8400 \left(\frac{\theta}{\beta} \right)^3 - 4200 \left(\frac{\theta}{\beta} \right)^4 \right]
 \end{aligned} \right.
 \end{aligned}$$

For Rise: $h = 50$, $\beta = 75^\circ = \frac{75}{180} \pi = \frac{5}{12} \pi \Rightarrow S_1, V_1, A_1, J_1$

For Dwell: $S_2 = h = 50$, $V_2 = 0$, $A_2 = 0$, $J_2 = 0$, $\theta \rightarrow \theta - \frac{75}{180} \pi$

For Fall: $S_3 = h_3 - S_2 \Rightarrow S_3, V_3, A_3, J_3$
 $h = 50$, $\beta = 75^\circ = \frac{5}{12} \pi$, $\theta \rightarrow \theta - \frac{150}{180} \pi$

For Dwell: $S_4 = 0$, $V_4 = 0$, $A_4 = 0$, $J_4 = 0$, $\theta \rightarrow \theta - \frac{225}{180} \pi$

Problem 8-9

This is a single dwell case

Using a polynomial with the following boundary conditions.

$$\left. \begin{array}{l} \theta=0, \quad s=0, \quad v=0, \quad a=0 \\ \theta=\beta_1, \quad s=h, \quad v=0 \\ \theta=\beta_2, \quad s=0, \quad v=0, \quad a=0 \end{array} \right\} \begin{array}{l} h = 2 \text{ in} \\ \beta_1 = 60^\circ \text{ deg} \\ \beta_2 = 150^\circ \text{ deg} \end{array}$$

(set $v=0$ for $\theta=\beta_1$, at the end of rise is to avoid overshoot of displacement)

$$\text{let } s = c_0 + c_1 \left(\frac{\theta}{\beta}\right) + c_2 \left(\frac{\theta}{\beta}\right)^2 + c_3 \left(\frac{\theta}{\beta}\right)^3 + c_4 \left(\frac{\theta}{\beta}\right)^4 + c_5 \left(\frac{\theta}{\beta}\right)^5 + c_6 \left(\frac{\theta}{\beta}\right)^6 + c_7 \left(\frac{\theta}{\beta}\right)^7 + c_8 \left(\frac{\theta}{\beta}\right)^8$$

$$\text{For } \theta=0, s=0, v=0, a=0 \Rightarrow c_0=0, c_1=0, c_2=0$$

$$c_3 A^3 + c_4 A^4 + c_5 A^5 + c_6 A^6 + c_7 A^7 = h \quad (1)$$

$$\text{For } \theta=\beta_1, v=0 \Rightarrow 3c_3 A^2 + 4c_4 A^3 + 5c_5 A^4 + 6c_6 A^5 + 7c_7 A^6 = 0 \quad (2)$$

$$s=h \quad (A = \beta_1/\beta = 60/150 = 0.4)$$

$$\text{For } \theta=\beta_2, s=0, v=0, a=0 \Rightarrow c_3 + c_4 + c_5 + c_6 + c_7 = 0 \quad (3)$$

$$3c_3 + 4c_4 + 5c_5 + 6c_6 + 7c_7 = 0 \quad (4)$$

$$6c_3 + 12c_4 + 20c_5 + 30c_6 + 42c_7 = 0 \quad (5)$$

$$\text{Solving (1), (2), (3), (4), (5)} \Rightarrow C \begin{bmatrix} c_3 \\ c_4 \\ c_5 \\ c_6 \\ c_7 \end{bmatrix} = \begin{bmatrix} h \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} A^3 & A^4 & A^5 & A^6 & A^7 \\ 3A^2 & 4A^3 & 5A^4 & 6A^5 & 7A^6 \\ 1 & 1 & 1 & 1 & 1 \\ 3 & 4 & 5 & 6 & 7 \\ 6 & 12 & 20 & 30 & 42 \end{bmatrix} \Rightarrow \begin{bmatrix} c_3 \\ c_4 \\ c_5 \\ c_6 \\ c_7 \end{bmatrix} = C^{-1} \begin{bmatrix} h \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow c_3 = 289.352, c_4 = -1229.745, c_5 = 1953.125, c_6 = -1374.421$$

$$c_7 = 361.690$$

$$\therefore s(\theta) = c_3 \left(\frac{\theta}{\beta}\right)^3 + c_4 \left(\frac{\theta}{\beta}\right)^4 + c_5 \left(\frac{\theta}{\beta}\right)^5 + c_6 \left(\frac{\theta}{\beta}\right)^6 + c_7 \left(\frac{\theta}{\beta}\right)^7$$

$$\beta = 150^\circ \text{ or } \frac{5\pi}{6}$$

Problem 8-8

Conditions:

$$\begin{cases} \text{Rise } \beta_1 = 45^\circ, h_1 = 1.5 \text{ in} \\ \text{Dwell } \beta_2 = 150^\circ, h_2 = 0 \text{ in} \\ \text{Fall } \beta_3 = 90^\circ, h_3 = 1.5 \text{ in} \\ \text{Dwell } \beta_4 = 75^\circ, h_4 = 0 \end{cases} \text{ Minimum velocity}$$

Using modified sinusoidal motion in SCCA for rise and fall
 $b = 0.25, c = 0, d = 0.75, Ca = 5.528$

Rise:

For $0 \leq x \leq \frac{b}{2}, y_1 = Ca \left(\frac{b}{\pi} x - \left(\frac{b}{\pi} \right)^2 \sin \left(\frac{\pi x}{b} \right) \right)$

For $\frac{b}{2} \leq x \leq \frac{1-d}{2}, y_2 = Ca \left(\frac{x^2}{2} + b \left(\frac{1}{\pi} - \frac{1}{2} \right) x + b^2 \left(\frac{1}{8} - \frac{1}{\pi^2} \right) \right)$

For $\frac{1-d}{2} \leq x \leq \frac{1+d}{2}, y_3 = Ca \left[\left(\frac{b}{\pi} + \frac{c}{2} \right) x + \left(\frac{d}{\pi} \right)^2 + b^2 \left(\frac{1}{8} - \frac{1}{\pi^2} \right) - \frac{(1-d)^2}{8} - \left(\frac{d}{\pi} \right)^2 \sin \left(\frac{\pi}{d} \left(x - \frac{1-d}{2} \right) \right) \right]$

For $\frac{1+d}{2} \leq x \leq 1 - \frac{b}{2}, y_4 = Ca \left[-\frac{x^2}{2} + \left(\frac{b}{\pi} + 1 - \frac{b}{2} \right) x + (2d^2 - b^2) \left(\frac{1}{\pi^2} - \frac{1}{8} \right) - \frac{1}{4} \right]$

For $1 - \frac{b}{2} \leq x \leq 1, y_5 = Ca \left[\frac{b}{\pi} x + \frac{2(d^2 - b^2)}{\pi^2} + \frac{(1-b)^2 - d^2}{4} - \left(\frac{b}{\pi} \right)^2 \sin \left(\frac{\pi}{b} (x-1) \right) \right]$

Fall: Do the same for $y_1, y_2 \dots y_5$. Note the interval is different

$$\begin{cases} \text{For } 0 \leq x' \leq \frac{b}{2}, y_1' = 1 - y_1 \\ \text{For } \frac{b}{2} \leq x' \leq \frac{1-d}{2}, y_2' = 1 - y_2 \\ \text{For } \frac{1-d}{2} \leq x' \leq \frac{1+d}{2}, y_3' = 1 - y_3 \\ \text{For } \frac{1+d}{2} \leq x' \leq 1 - \frac{b}{2}, y_4' = 1 - y_4 \\ \text{For } 1 - \frac{b}{2} \leq x' \leq 1, y_5' = 1 - y_5 \end{cases}$$

All are in local coordinates.

In the rise, $0 \leq \theta \leq \beta_1, x = \frac{s}{\beta_1}, y = \frac{s}{h_1}$
 In the fall, $0 \leq \theta \leq \beta_3, x' = \frac{s}{\beta_3}, y' = \frac{s}{h_3}$

Substituting them in the equations, the displacement functions are derived.

For each dwell segment, the displacement is that at the end of the previous segment and remains as constant.